Exp-02 STUDY OF DIFFERENT TYPES OF NETWORK CABLES

16/7/26

Aim

To study and understand different types of network cable, To make your own ethernet cross over cable / straight cable.

a) Understand different types of network cable different types of cable used in network and

① Unshielded twisted Pair (UTP) cables.
② Shielded Twisted Pair (STP) Cables.
③ Coaxial cable
④ Fibre optic cables

CABLE TYPE	CATEGORY	MAXIMUM DATA TRANSMISSI	ADVANTAGES / DISADVANTAGES	APPLICATION
UTP	Category 3	10 bps	Advantages • Cheap in cost to fibre as they have smaller diameter Disadvantage . More prone to EMI and noise	to Base-T Ethernet
	Category 5	Up to 100 Mbps		Fast Ethernet Gigabit Ethern
	Category 5e	1 Gbps		
STP	Category 6, 6a	10 Gbps	Advantage • Shielded • Faster than UTP . Less susceptible to noise and maintenance Disadvantage • Expensive • Heavy	Gigabit Ethern 10a Ethernet (same) widely used in Europe
SSTP	Category 7	10 Gbps		Gigabit Ethe 10G, Ethernet Cloud


Clip underneath, pins facing you
Clip underneath, pins facing you
green solid
white/brown stripe
A
B

Coaxial cable	RG-6 RG-59 RG-11	10-100 Mbps	• High bandwidth • Immune to interference • Low loss bandwidth • Reliable	Speed of signal is 500K Telephone network high speed internet connections
fiber optical cable	Single mode multi mode	100 Gbps	ADVANTAGES • HIGH SPEED • HIGH BANDWIDTH • HIGH SECURITY • LONG DISTANCE DISADVANTAGES • EXPENSIVE • REQUIRED SKILL INSTALLED	Maximum distance of fiber optic cable is around 100 miles

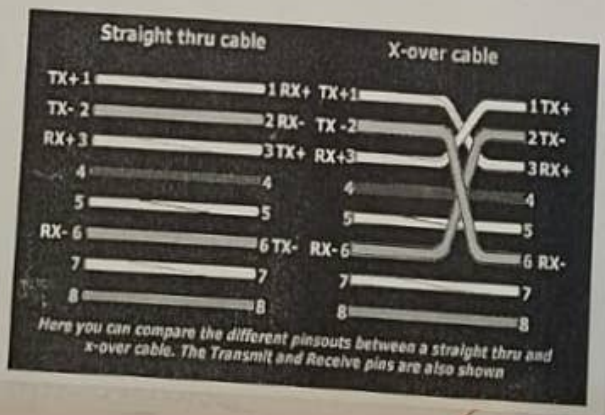
Make your own Ethernet cross-over CABLE / STRAIGHT cable:

TOOLS AND PARTS NEEDED:

* Ethernet cabling. CAT5E is rated for gigabit support, but cat5 cabling could work, just over shorter distances
* A crimping tools. This is an all in one networking tool shapes to bend down the pins in the plug and strip and cut the shielding off the cable
  ① Two RJ45 plugs.
  ⑪ Optional two plugs shields

Straight through network cable: both sides should be A
Crossover cable: One side A, one side B





Here you can compare the different pinsouts between a straight thru and x-over cable. The Transmit and Receive pins are also shown

Steps:

① To start construction of the drill, begin by threading shields onto the cable.

② Next, strip approximately 1.5 cm of cable shielding from both ends. The crimping tool has a mould and to compress the tank

③ After you will be used to untangle the wire there should be four "twisted pair". Untwist each to start, untangle them from top to bottom. One end should be in untangled A and the other in B

④ Straighten the wires. Trim them evenly and wire the outer connect. Hold the RJ45 plug with clip away and gold pins facing you. Push wire in carefully without untwisting the sections.

⑤ Next, push the cable right in. The notch over the cable shielding and if it isn't that mean that you stripped off too much shielding simply strip the cable back a little more.

⑥ After the wires are neatly sitting inside the plug, insert it into the crimping tool and push down. It should connect but pushing too hard can crack the fragile plastic plug.


A
underneath,

(VII) Jainy, minat the other end wing diagr
B to main a crossover cable, wing diagr
A to main a straight through cable.

To test it. plug into and attempt to
connect two devices directly.

Result: This different types of network
cable were studied successfully and straight
through and cross over ethernet cables
were prepared and tested successfully.